

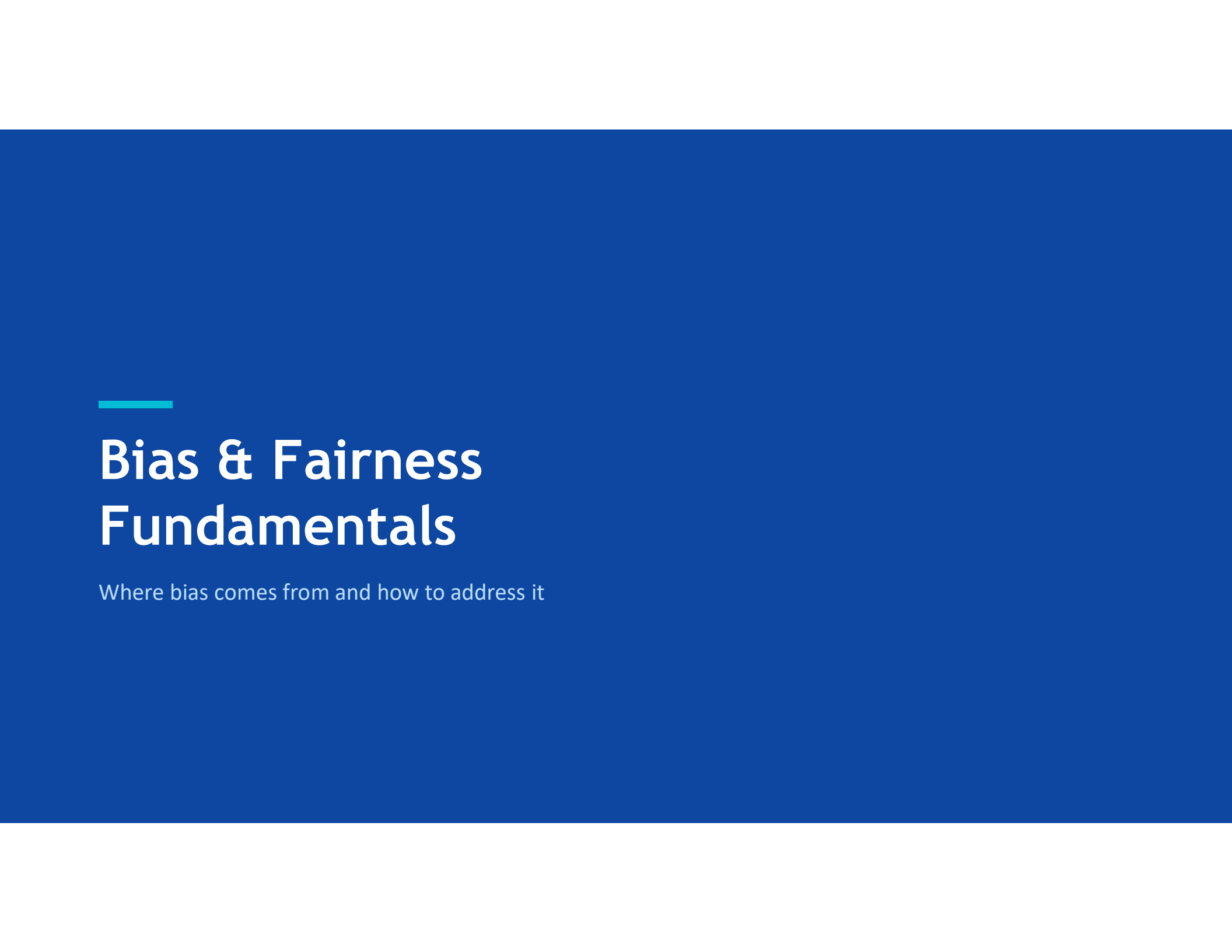
Responsible AI & Fine-Tuning Intro

Building AI that is fair, safe, transparent, and customizable

Freshers GenAI Training Program | FIL

Learning Objectives

- Understand bias and fairness fundamentals — how bias enters AI systems and how to detect it
- Detect and mitigate hallucinations in LLM outputs using practical techniques
- Apply security and privacy best practices for GenAI applications in enterprise environments
- Grasp compliance and policy awareness basics relevant to AI deployment
- Understand when fine-tuning is appropriate vs. prompting and RAG alternatives
- Learn PEFT and LoRA concepts — efficient fine-tuning without retraining the full model



Bias & Fairness Fundamentals

Where bias comes from and how to address it

How Bias Enters AI Systems

- Training data bias: models learn from internet text — which reflects historical and societal biases
- Representation bias: underrepresented groups in training data → worse performance for those groups
- Label bias: human annotators bring their own biases into labeled datasets used for fine-tuning
- Selection bias: choosing which data to include/exclude shapes the model's worldview
- Measurement bias: proxies used for sensitive attributes (e.g., ZIP code as proxy for race)
- Amplification: models don't just reflect biases — they can amplify them through confident, authoritative outputs

Types of Fairness in AI

GROUP FAIRNESS

- Demographic parity: outcomes should be distributed equally across groups
- Equal opportunity: equal true positive rates across groups
- Predictive parity: equal precision across groups
- Tension: these definitions often conflict with each other — no single 'fair' solution exists
- Context matters: which fairness metric to prioritize depends on the use case and stakeholders

INDIVIDUAL FAIRNESS

- Similar individuals should receive similar treatment — regardless of group membership
- Harder to define: what makes two individuals 'similar' requires domain expertise
- Counterfactual fairness: would the outcome change if the person's protected attribute were different?
- In LLMs: does the model's response change based on names, pronouns, or cultural markers?
- Testing: systematically swap demographic indicators in prompts and compare outputs

Detecting & Mitigating Bias in LLM Applications

- Red-teaming: systematically probe the model with adversarial inputs targeting bias patterns
- Prompt auditing: test identical queries with different demographic markers — 'John' vs. 'Jamila'
- Output analysis: monitor response quality, sentiment, and helpfulness across user segments
- Debiasing prompts: instruct the model to be fair, consider multiple perspectives, avoid stereotypes
- Diverse evaluation: include evaluators from varied backgrounds in your quality assessment process
- Feedback loops: build mechanisms for users to flag biased outputs — and actually act on the data



Hallucination Detection & Mitigation

When AI makes things up — and how to catch it

The Hallucination Problem

15-25%

Factual Error Rate

Without grounding, LLMs produce plausible-sounding but incorrect facts in 15-25% of responses

RAG

Primary Mitigation

Retrieval-augmented generation reduces hallucination by grounding answers in real documents

Chain

Verification Chains

Multi-step verification: generate → verify against sources → flag unsupported claims

Hallucination Types & Detection

TYPES OF HALLUCINATION

- Factual hallucination: states incorrect facts confidently — wrong dates, invented statistics
- Fabricated citations: invents paper titles, authors, DOIs that don't exist
- Conflation: mixes up facts between similar entities — attributes one person's work to another
- Extrapolation: goes beyond source material and adds unsupported conclusions
- Formatting hallucination: generates syntactically valid but semantically wrong code/JSON

DETECTION TECHNIQUES

- Source verification: compare claims against retrieved documents — does the source actually support this?
- Consistency checking: ask the same question multiple ways — inconsistent answers signal uncertainty
- Confidence scoring: ask the model to rate its confidence — low confidence correlates with hallucination
- Fact-checking chains: use a second LLM call to verify claims from the first response
- Human review: for high-stakes outputs, implement human-in-the-loop verification workflows

Practical Hallucination Mitigation Strategies

- Strategy 1: RAG — always ground responses in retrieved documents, never rely solely on parametric knowledge
- Strategy 2: Citation enforcement — instruct the model to cite specific sources for each claim
- Strategy 3: Temperature control — use low temperature (0.1–0.3) for factual tasks to reduce randomness
- Strategy 4: Output constraints — structured output (JSON, enums) limits the space for fabrication
- Strategy 5: Abstention — train/prompt the model to say 'I don't know' when uncertain
- Strategy 6: Multi-model consensus — cross-check outputs from different models, flag disagreements



Security & Privacy for GenAI Apps

Protecting data and preventing attacks

Security Threats Specific to GenAI

- Prompt injection: malicious inputs that override system instructions — the #1 GenAI security risk
- Data leakage: LLMs can memorize and regurgitate training data — including PII and sensitive information
- Model inversion: attackers extract training data by querying the model with carefully crafted inputs
- Supply chain attacks: compromised prompt templates, malicious dependencies, poisoned fine-tuning data
- Side-channel attacks: inference timing, token probabilities can reveal information about the system
- Social engineering: users trick the model into revealing system prompts, tool definitions, or internal data

Security & Privacy Best Practices

INPUT SECURITY

- Input sanitization: strip/escape special characters and known injection patterns
- Input length limits: prevent token-based attacks with reasonable size caps
- Content filtering: detect and block harmful, offensive, or policy-violating inputs pre-model
- Rate limiting: prevent brute-force attacks and resource exhaustion per user/IP
- Authentication: every GenAI endpoint requires proper auth — no anonymous LLM access

DATA PRIVACY

- PII detection: scan inputs and outputs for personal data (names, emails, SSNs) — redact before processing
- Data residency: ensure data stays in approved regions — check your model provider's data processing terms
- Logging hygiene: never log full prompts containing user PII — log sanitized versions only
- Consent: users must know their data is being processed by an AI — transparency is mandatory
- Retention policies: define how long conversation data is stored — implement auto-deletion

Compliance & Policy Awareness

- GDPR (EU): right to explanation, data portability, right to deletion — applies to AI decisions about EU residents
- AI Act (EU): risk-based framework — high-risk AI requires transparency, human oversight, and documentation
- FIL-specific policies: internal AI acceptable use policy, data classification, model governance requirements
- Model cards: document model capabilities, limitations, intended use, and known biases
- Audit trails: maintain logs of AI decisions for regulatory review — especially in financial services
- Human override: for high-impact decisions, ensure a human can always review and override the AI's output



Fine-Tuning: When & How to Customize Models

Adapting foundation models to your specific needs

When to Fine-Tune vs. Use Alternatives

FINE-TUNE WHEN...

- You need consistent output format that prompting can't reliably achieve
- Domain-specific language/jargon that the base model handles poorly
- You have 1000+ high-quality labeled examples for the task
- You need to reduce token usage — a fine-tuned small model can replace a prompted large model
- Latency requirements: fine-tuned small model is faster than prompted large model

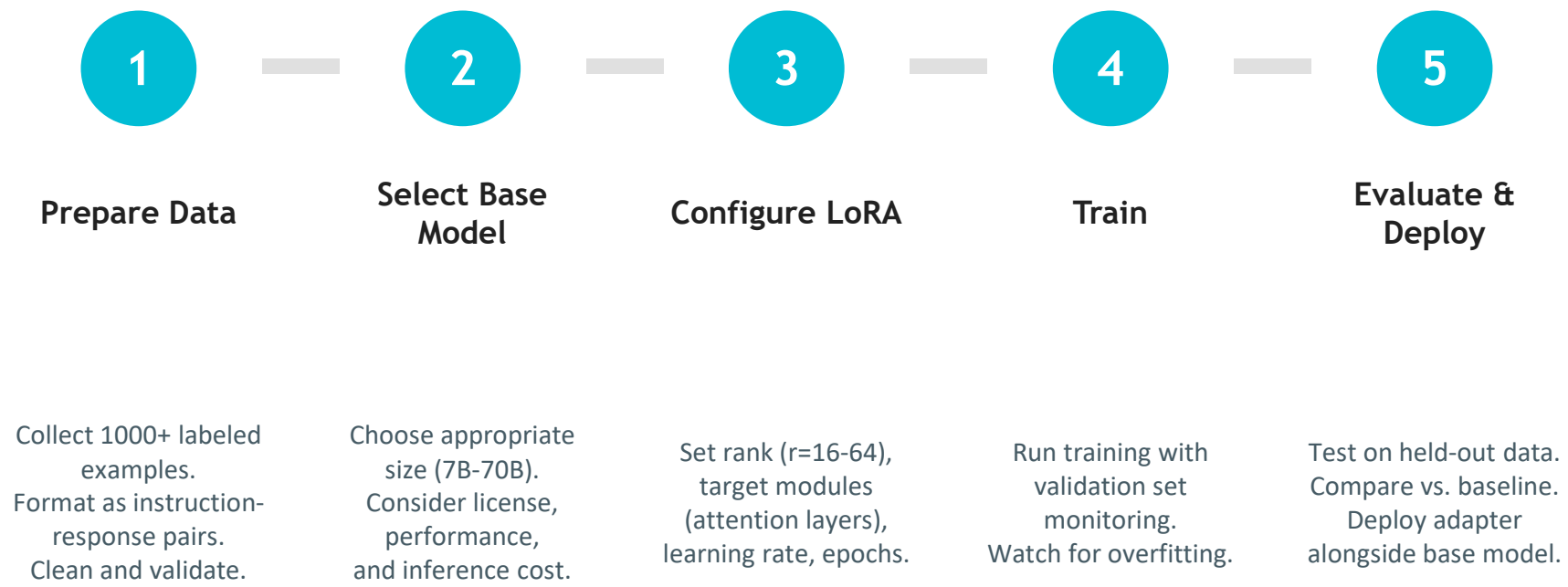
DON'T FINE-TUNE WHEN...

- Prompting or few-shot examples already achieve acceptable quality
- RAG can provide the domain knowledge the model needs
- You have fewer than 100 training examples — too little data for meaningful adaptation
- The task requires up-to-date information — fine-tuning freezes knowledge at training time
- You need to change behavior frequently — prompt changes are instant, fine-tuning takes hours/days

PEFT & LoRA: Efficient Fine-Tuning

- Full fine-tuning: update ALL parameters (7B model = 7 billion weights) — expensive, slow, needs many GPUs
- PEFT (Parameter-Efficient Fine-Tuning): update only a small fraction of parameters — 0.1% to 1% of the model
- LoRA (Low-Rank Adaptation): the most popular PEFT technique — adds small trainable matrices alongside frozen weights
- How LoRA works: instead of updating a 4096×4096 weight matrix, train two small matrices (4096×16 and 16×4096)
- Benefits: 10-100x less GPU memory, trains in hours instead of days, produces a tiny adapter file (~100MB)
- QLoRA: combines LoRA with 4-bit quantization — fine-tune a 70B model on a single GPU!

Fine-Tuning Workflow



HANDS-ON EXERCISES

1

Ethical Scenario Review

Analyze 3 real-world AI bias/fairness case studies. For each: identify the bias, its source, and a mitigation plan

2

Fine-Tuning Data Preparation

Curate a starter dataset (50–100 examples) for a domain-specific task, formatted as instruction-response pairs

DELIVERABLE: Capstone Milestone 2 — Responsible AI checklist completed + evaluation criteria for your capstone project

KEY TAKEAWAYS

- 01** Bias is systemic, not accidental — it enters through data, labels, design, and deployment decisions
- 02** Hallucination mitigation: RAG + citations + low temperature + structured output + abstention
- 03** Security: prompt injection is the SQL injection of GenAI — sanitize inputs, constrain outputs
- 04** Compliance awareness: GDPR, AI Act, and internal policies govern what you can build and deploy
- 05** Fine-tuning decision: prompting + RAG first, fine-tune only when data quality and quantity justify it

NEXT WEEK: Advanced Trends + Capstone Showcase — multimodal AI, MLOps, and your team's final presentation